An orbit is continuous falling

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain a circular orbit with gravity and sideways motion.

SUCCESS CRITERIA - I can...

1. Describe orbit as free fall rather than absent gravity.
2. Distinguish tangent velocity from inward force.
3. Connect circular speed to orbital radius.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “orbit” and “gravity” to explain your model or result. Sentence starters are optional.

orbit: a path around another body under gravity

gravity: an attractive interaction between masses

radius: distance from the center to the edge of a circle or sphere

01 Bring back an earlier idea

A projectile launches horizontally at 5 m/s from height 20 m.
Use g=10 and no drag.
Find flight time, range and final vertical velocity.

Word help: vertical = up and down, usually the y direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $GM/r^2 = v^2/r$; $v = \sqrt{(GM/r)}$.
Circular orbit outside a spherical body of mass M; r is orbital radius. Neglect atmosphere and other bodies.

Worked reasoning: A toy circular orbit has radius 4 m and speed 2 m/s. Its period is circumference/speed=$2\pi \times 4/2 = 4\pi$ s. Gravity continuously changes the velocity direction.

Your turn: Draw a quarter-orbit with tangent velocity arrows and inward gravitational arrows.

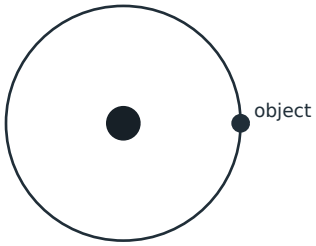
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Draw the path of an orbiting object and its velocity/force arrows.
Explain how it can fall without immediately hitting the surface.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

04 Solve and justify

For a toy planet GM=400 m^3/s^2 and circular radius 16 m, find orbital speed, period and acceleration.

Sentence starter: My design shows ... . It meets the requirement by ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Why does a rocket not need to keep firing forward to remain in an ideal circular orbit?

Word help: ideal = using stated simplifying conditions

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A 4 kg object moves at 6 m/s in an ideal circle under a force perpendicular to velocity.
Explain which parts of velocity change and whether the force does work.

Word help: ideal = using stated simplifying conditions

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

If circular orbital radius increases by factor 4 around same body, compare speed and period.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Around one source, a circular orbit radius increases ninefold.
Compare speed, period and gravitational acceleration without inserting a source mass.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3